Callbacks

A callback is a function that's invoked from somewhere else. You see, normally you invoke a function yourself, whenever you feel the need:

```
partyMessage()
# It's party time!
```

A function used as a callback is different. Instead of invoking the function, imagine giving it to Scruffy and asking him to invoke it later. *That's a callback*. Similarly, when you invoke `setTimeout` with `partyMessage`, you're telling `setTimeout` to invoke it later. You're using the `partyMessage` function as a callback.

These callback functions have a very important feature—they're *asynchronous*. What does that mean? To find out, try `setTimeout partyMessage` on the REPL again, but this time make the duration 5 seconds and enter the expression `1+1` before you see the party message appear. Be quick!

```
setTimeout partyMessage, 5000
1 + 1
# 2

# ... 5 seconds later
# It's party time!
```

Notice how the REPL evaluated the expression `1+1` before the timeout was complete? That's why asynchronous callbacks are important; you can go off and do other things while you're waiting for the response. That's exactly what Agtron does in figure 3.2 as he waits for Scruffy to call back.

Figure 3.2. Callbacks are asynchronous.